

A 10/3 asymptotic lower coefficient through partite saving and linearization

Research continuation, 7 October 2026. This is a continuation of the repository's partite cover-number programme, not a separate claim of resolving Ryser's conjecture. All combinatorial arguments are written below. Two attributed external inputs are identified explicitly.

1. The result and its scope

Let H be a finite simple intersecting r -partite r -uniform hypergraph, $r \geq 2$, with m distinct edges and cover number $t = \tau(H)$. Edges meet every part in exactly one vertex. Empty hypergraphs have cover number zero, and isolated vertices are irrelevant.

Theorem 1. For every $\delta > 0$ there is a finite constant C_δ , independent of r, H , such that

$$m \geq 5\tau(H) - \left(\frac{5}{3} + \delta\right)r - C_\delta. \quad (1)$$

Consequently, whenever the defining class is nonempty,

$$f(r) = \min_{H: \tau(H) \geq r-1} |E(H)| \geq \left(\frac{10}{3} - \delta\right)r - C_\delta - 5. \quad (2)$$

In particular, for every $\varepsilon > 0$ and sufficiently large r , $f(r) \geq (10/3 - \varepsilon)r$.

The preceding complete elementary note proves the finite bound $f(r) \geq \lceil 13r/4 - 10 \rceil$. The new coefficient 10/3 is greater than 13/4, and also greater than the directly applicable 293/96 coefficient of Aharoni–Barát–Wanless [ABW]. This note does not supply numerical thresholds from Kahn's theorem, determine $f(r)$, prove that its defining class is nonempty for every rank, or settle the inequality $\tau(H) \leq r - 1$. No priority claim is made.

The new ingredient is the saving-sensitive linearization estimate $e \geq \binom{q}{2} - qr/2 - 3W$, coupled to a cover saving of W/r . Degree-five peeling and the matching/star strategy follow Sivashankar [S, Section 4]. The inherited degree-three lemma is proved in the appendix. The other input is the published small-codegree edge-colouring theorem of Kahn, precisely stated in Section 2. It is not re-proved or Lean-formalized here. The theorem is unconditional as a deduction from that standard theorem, with the dependency made explicit.

2. The edge-colouring input

For an auxiliary hypergraph, let D be its maximum vertex degree, and let its maximum **codegree** be the greatest number of edges containing any specified pair of distinct vertices. Its chromatic index is the minimum number of colours needed to colour edges so that each colour class is a matching.

A hypergraph is **linear** if two distinct edges meet in at most one vertex, equivalently its maximum codegree is at most one for a simple hypergraph.

Input K (Kahn, rank-four consequence). For every $\eta > 0$ there is an integer $D_0 \geq 1$ such that a linear four-uniform hypergraph of maximum degree $D \geq D_0$ has chromatic index at most $(1 + \eta)D$. Hence it has a matching of size at least $e/((1 + \eta)D)$, where e is its edge count.

This follows from [KKKMO, Theorem 3.1], attributed there to Kahn [K]: for fixed rank k and $\eta > 0$, there is $\gamma > 0$ such that rank at most k , maximum degree at most D , and maximum codegree at most γD imply list chromatic index at most $(1 + \eta)D$. Set $k = 4$ and increase the integer threshold so that $1 \leq \gamma D$. Ordinary colouring follows from list colouring; the largest colour class gives the stated matching. All uses below verify uniformity, linearity and the degree threshold.

3. Peeling and the two inherited estimates

Select vertices of current degree at least five and delete their incident edges until the residual J has maximum degree at most four. Let k count selected vertices and $q = |E(J)|$. As in the preceding note,

$$m \geq q + 5k, \quad t \leq k + \tau(J), \quad m \geq q + 5(t - \tau(J)). \quad (3)$$

For $q > 0$, fix an edge of J and count its intersections with the others. Since its r vertices have degrees at most four, $q - 1 \leq 3r$. For an empty residual the same inequality holds. Thus

$$q \leq 3r + 1. \quad (4)$$

The degree-three lemma in the appendix implies

$$\tau(J) \leq (q + r + 4)/4. \quad (5)$$

Indeed, choose a maximal matching of four-point incidence blocks defined by degree-four vertices of J , say a blocks. The corresponding a vertices cover $4a$ edges; the remaining hypergraph has maximum degree at most three by maximality. Applying the appendix to its $q - 4a$ edges gives $a + (q - 4a + r + 4)/4 = (q + r + 4)/4$. It remains r -uniform. This argument also applies to an empty remaining hypergraph. From (3)–(5),

$$m \geq 5t - 5r/4 - q/4 - 5. \quad (6)$$

If $q \leq 5r/3$, this already gives $m \geq 5t - 5r/3 - 5$. The stronger analysis is needed only for $q \geq 5r/3$.

4. A saving-sensitive count of linear four-blocks

Let x_i count vertices of residual degree i , $1 \leq i \leq 4$, and put $W = x_3/2 + x_4$. Set

$$S = \sum_v \binom{d_J(v)}{2} = x_2 + 3x_3 + 6x_4.$$

Since every edge pair intersects, the total intersection excess is

$$T = S - \binom{q}{2} = \sum_{\{A, B\} \subset E(J)} (|A \cap B| - 1) \geq 0. \quad (7)$$

Incidence counting gives $x_1 + 2x_2 + 3x_3 + 4x_4 = qr$.

Form the auxiliary four-uniform multihypergraph Q on the q edge-points of J : each degree-four vertex of J supplies one four-point block. Retain repeated blocks initially. Let λ_{AB} be the number of those blocks containing edge-points A, B . Then $\lambda_{AB} \leq |A \cap B|$, so

$$X := \sum_{\{A,B\}} (\lambda_{AB} - 1)_+ \leq T. \quad (8)$$

Whenever some pair has current codegree at least two, delete one block containing it. This lowers the integer X by at least one and cannot increase any pair excess. After at most the original X deletions, the retained four-uniform hypergraph Q' is linear. It is also simple: two repeated four-blocks would have codegree at least two. Write $e = |E(Q')|$. The exact algebra gives

$$\begin{aligned} e \geq x_4 - T &= \binom{q}{2} - qr/2 - 3W + x_1/2 \\ &\geq \binom{q}{2} - qr/2 - 3W. \end{aligned} \quad (9)$$

The lower bounds may be negative; no positivity is assumed here.

Separately, selecting every degree-three and degree-four vertex of one part covers disjoint edge sets. If their counts in part c are a_c, b_c , pairing the remaining edges gives

$$\tau(J) \leq a_c + b_c + \lceil (q - 3a_c - 4b_c)/2 \rceil \leq q/2 + 1/2 - a_c/2 - b_c.$$

The part savings have total W , so one is at least W/r . Define a shifted cover saving

$$s = q/2 + 1 - \tau(J).$$

The pairing cover shows $s \geq 0$, and the part cover gives

$$W \leq rs. \quad (10)$$

5. A uniform matching/star bound, including small degrees

Fix $\eta > 0$ and take D_0 from Input K. Let D be the maximum degree of Q' , taking $D = 0$ if it is empty.

If $D \geq D_0$, Input K gives a matching of at least $e/((1 + \eta)D)$ blocks. Their original defining vertices cover four edges each, and pairing the rest gives

$$s \geq e/((1 + \eta)D).$$

For a second cover, take the D blocks through a maximum-degree edge-point. Their other points are disjoint by linearity, so they cover exactly $1 + 3D$ original edges. The corresponding D vertices together with a pairing of the remaining edges give

$$\tau(J) \leq D + \lceil (q - 1 - 3D)/2 \rceil \leq q/2 - D/2, \quad s \geq D/2.$$

Multiplication of these nonnegative bounds yields $e \leq 2(1 + \eta)s^2$.

If $D < D_0$, incidence counting in Q' instead gives $4e \leq qD < qD_0$. Consequently in both cases

$$e \leq 2(1 + \eta)s^2 + qD_0/4. \quad (11)$$

This explicit small-degree case avoids applying an asymptotic theorem outside its threshold.

Combining (9)–(11) gives, with $B = 1/2 + D_0/4$,

$$2(1 + \eta)s^2 + 3rs \geq q(q - r)/2 - qB. \quad (12)$$

6. The 5/3 transition and all error terms

Assume $q \geq 5r/3$, and set

$$b = (1 + \eta)s, \quad \ell = 3q/10 - r/3, \quad E = qB/(3r).$$

All three are nonnegative. Since $\eta, s, r \geq 0$, (12) implies

$$2b^2 + 3rb \geq q(q - r)/2 - 3rE. \quad (13)$$

The exact factorization

$$q(q - r)/2 - (2\ell^2 + 3r\ell) = \frac{(3q - 5r)(24q - 35r)}{225} \geq 0 \quad (14)$$

is valid in this range: the second factor is at least $5r$. It follows that

$$b + E \geq \ell. \quad (15)$$

For if $b + E < \ell$, then $\ell - b > E$ and

$$(2\ell^2 + 3r\ell) - (2b^2 + 3rb) = (\ell - b)(2(\ell + b) + 3r) > 3rE,$$

contradicting (13)–(14).

From (4) and $r \geq 2$, $q \leq 7r/2$. Therefore

$$0 \leq \ell \leq r, \quad 0 \leq E \leq C := 7B/6. \quad (16)$$

Equations (15)–(16) imply

$$s \geq \ell - \eta r - C. \quad (17)$$

If $s > r$, this is immediate. If $s \leq r$, use $(1 + \eta)s + E \leq s + \eta r + C$ in (15).

Finally (3) and the definition of s give

$$m \geq 5t - 3q/2 + 5s - 5 \geq 5t - (5/3 + 5\eta)r - 5C - 5. \quad (18)$$

The q terms cancel exactly since $5\ell = 3q/2 - 5r/3$. For $q \leq 5r/3$, (6) gives a stronger bound than (18). Taking $\eta = \delta/5$ proves Theorem 1 with

$$C_\delta = 5 + \frac{35}{6} \left(\frac{1}{2} + \frac{D_0(\delta/5)}{4} \right). \quad (19)$$

To obtain the final ε formulation, take $\delta = \varepsilon/2$ and then choose r so large that $(C_\delta + 5)/r \leq \varepsilon/2$. This threshold is existential, matching the nature of Input K. $\square$

7. Verification limits and next structural problem

The proof handles arbitrary part widths, repeated pair intersections, empty residuals, and the small-degree alternative in the auxiliary graph. Deleting repeated original edges preserves all hypotheses and the cover number, so the bounds also apply to the number of distinct edges in a hypergraph originally presented with multiplicities.

The exact checker verifies the identities coefficientwise and checks the linearization procedure and its inequality on all intersecting binary-width families in ranks two and three, and selected larger exact

examples. These are diagnostics. Lean checks the scalar transitions with explicit hypotheses; it does not formalize the combinatorics or Input K. The written proof, together with that precisely cited standard theorem, is the proof of the universal result. No numerical experiment supplies a general step.

The transition here is $q/r = 5/3$. A further improvement requires beating the simultaneous cover savings W/r and the linearized matching/star bound. The method retains a precise obstruction: large intersection excess destroys many four-blocks, whereas larger W already saves cover vertices through the parts. Investigating which extremal degree distributions and intersection patterns are actually realizable is the next substantive step.

References

- [ABW] R. Aharoni, J. Barát and I. M. Wanless, *Multipartite hypergraphs achieving equality in Ryser’s conjecture*, Graphs and Combinatorics 32 (2016), 1–15, DOI:10.1007/s00373-015-1575-9, arXiv:1409.4833v2.
- [S] V. Sivashankar, *An Improved Lower Bound for the Erdős–Lovász Cover Number Problem*, arXiv:2606.24878v2. Its Section 4 supplies the peeling, linearization philosophy and matching/star combination for the nonpartite class. Its Lemma 2 is restated with proof in the appendix below. The saving-sensitive estimate (9) and its coupling to the part cover are the contribution of the present continuation.
- [K] J. Kahn, *Asymptotically good list-colorings*, Journal of Combinatorial Theory, Series A 73 (1996), 1–59.
- [KKKMO] D. Y. Kang, T. Kelly, D. Kühn, A. Methuku and D. Osthus, *Solution to a problem of Erdős on the chromatic index of hypergraphs with bounded codegree*, Proceedings of the London Mathematical Society 129 (2024), no. 6, e70011, arXiv:2110.06181. Theorem 3.1 states Input K’s parent theorem. The statement was checked in the primary manuscript, including its rank and codegree hypotheses.

Appendix. Full proof of the inherited degree-three lemma

Lemma A (Sivashankar [S, Lemma 2]). For an intersecting r -uniform hypergraph L with q edges and maximum degree at most three,

$$4\tau(L) \leq q + r + 4. \quad (15)$$

The proof below restates its maximum-matching argument in our notation; this lemma and its local coloured-graph claim are not claimed as our contribution.

Proof. The empty case is immediate. Fixing one edge and counting its intersections with all other edges gives

$$q - 1 \leq \sum_{v \in e} (d(v) - 1) \leq 2r, \quad q \leq 2r + 1. \quad (16)$$

Treat the q edges as points. Each original vertex defines the block of points containing it, of size at most three. Every point pair is contained in a block. Take a maximum matching of the distinct three-point blocks; let its size be s , its covered points C , and its unmatched points U , with $|U| = u$. Thus $q = 3s + u$. There is no three-block inside U . The defining vertices of the matching, followed by a pairing of U , give a cover of size at most $s + \lceil u/2 \rceil$. It suffices to show

$$r \geq u + s - 2. \quad (17)$$

Put $\alpha = r - u - s + 2$, an integer. If $u = 0$, (16) gives $\alpha \geq (r + 5)/3 > 0$. Assume $u > 0$.

Select one witness vertex for each pair in U . These witnesses are distinct: witnesses for disjoint pairs cannot coincide because the maximum degree is three; witnesses for overlapping pairs cannot coincide because that would create a three-block inside U . Each selected witness contributes exactly two incidences with U ; its block is the selected pair, possibly together with one point of C . The witnesses contribute $u(u - 1)$ such incidences in total.

We need the following coloured-graph observation, setting $K = \max\{12, u + 4\}$. In a simple graph on u vertices, colour each edge with one of at most three colours. If two disjoint edges never have different colours, the sum of the three colour support sizes is at most K . Here a colour support consists of the vertices incident with edges of that colour.

To prove the observation, suppose first that colour one contains two disjoint edges e, f , and put $W = e \cup f$. Every edge of the other colours must meet both, so its endpoints belong to W . If no other colour is used, the assertion holds. If their combined edge family has no common vertex, each colour-one edge must also have both endpoints in W : an edge with an endpoint outside W would force its other endpoint to be common to that family. All three support sizes are then at most four, giving twelve. Otherwise the combined family has a common vertex $c \in W$. Its edges must join c to one of the two endpoints of the edge among e, f not containing c . There are at most two such edges, and simplicity makes their combined colour-support sum at most four. Adding the first support gives $u + 4$.

Now assume each colour class is pairwise intersecting. A simple pairwise-intersecting graph is a star or is contained in a triangle: take two edges ab, ac ; any edge avoiding a must be bc , which in turn forces every edge into that triangle. If all supports have size at most four, their sum is at most twelve. Otherwise one class is a star with at least four leaves. Every edge of the other colours must contain its centre, since two endpoints cannot meet four distinct leaves. Simplicity ensures each noncentral vertex belongs to at most one support. The centre belongs to at most three, giving a sum at most $u + 2$. This proves the observation.

For each matching triple M_i , form a coloured graph on U : the selected pair xy is given colour $z \in M_i$ when its witness block is $\{x, y, z\}$. Two disjoint pairs with different colours would allow replacing M_i by two disjoint three-blocks, increasing the matching. Thus the observation applies. The colour support sum counts the distinct pairs between U and M_i covered by selected witnesses, so at most K such pairs are covered. Across the s triples, at least $3su - sK$ of the U - C pairs are uncovered by selected witnesses.

For a nonselected vertex let a count its block points in U and c those in C . Since $a + c \leq 3$, one has $ac \leq 2a$. All remaining pairs have nonselected witnesses, so their incidence contribution to U is at least $(3su - sK)/2$. Counting the total ur incidences with U yields

$$ur \geq u(u - 1) + \frac{3su - sK}{2}. \quad (18)$$

This reasoning remains valid if the displayed lower bound is negative.

Equation (16) with $r = u + s - 2 + \alpha$ gives $s \leq u - 3 + 2\alpha$. Dividing (18) by $u > 0$ gives

$$\alpha \geq 1 - \frac{(K - u)s}{2u}. \quad (19)$$

If $\alpha < 0$, its integrality implies $\alpha \leq -1$ and $0 \leq s \leq u - 5$, hence $u \geq 5$. As $K - u \geq 0$, (19) implies $\alpha \geq 1 - (K - u)(u - 5)/(2u)$. This is greater than -1 . Indeed, for $5 \leq u \leq 8$ one has $K = 12$ and

$$4u - (12 - u)(u - 5) = u^2 - 13u + 60 = (u - 13/2)^2 + 71/4 > 0.$$

For $u \geq 8$, one has $K = u + 4$, giving $-1 + 10/u > -1$. This contradiction proves $\alpha \geq 0$, hence (17).
 Finally

$$4\tau(L) \leq 4s + 4\lceil u/2 \rceil \leq 4s + 2u + 2 \leq q + r + 4.$$

The lemma follows. $\square$